



Multi-agent task and motion planning trends analysis: a survey

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Abstract

Multi-agent Task and Motion Planning (TAMP) has become a key research frontier in autonomous agents, supported by the rapid evolution of Embodied AI and the increasing deployment of robot teams in real-world environments. Unlike single-agent planning, multi-agent TAMP must jointly reason about discrete task allocation, continuous motion feasibility, communication constraints, and execution uncertainties. This survey provides a systematic analysis of recent advances in multi-agent TAMP from 2021 to 2025. We first integrate and refine existing taxonomies, and propose a unified classification framework based on three dimensions. A total of 60 representative works are reviewed under this taxonomy, revealing dominant solution strategies. Furthermore, this survey summarizes emerging research trends, including LLM-based TAMP, decentralized adaptive planning frameworks, and reinforcement learning based TAMP solutions. Finally, we highlight key future directions, such as VLM-based planning, Theory-based planning, data privacy and long-horizon task planning.

Keywords Task and motion planning · Multi-agent system · AI · Autonomous system

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1 Introduction

Autonomous agents refer to systems capable of performing tasks, making decisions, and taking actions independently to achieve their goals with minimal human intervention (Natarajan et al. 2025). These agents appear in diverse forms, ranging from humanoid robots and robotic manipulators (Sandakalum and Ang 2022) to autonomous unmanned aerial vehicles (UAVs) (Jones et al. 2023), unmanned ground vehicles (UGVs) (Wang et al. 2024), and underwater vehicles (Guo et al. 2021). In real-world scenarios, autonomous agents have already become widely deployed, such as robotic arms in large-scale logistics warehouses, autonomous delivery vehicles in urban areas (Ćirović et al. 2014), harvesting manipulators in smart agriculture (Ji et al. 2023), and service robots in hotels (Choi et al. 2020). Numerous reviews have examined how various types of robots achieve autonomy from different technical perspectives (Dinelli et al. 2023; Bai et al. 2023; Chebab et al. 2015).

With the rapid development of artificial intelligence (AI), the concept of autonomous agents has further evolved into Embodied AI, which refers to AI systems integrated into physical robotic platforms (Natarajan et al. 2025). Typical embodiments include autonomous vehicles (Chen et al. 2025), autonomous drones (Natarajan et al. 2025), dexterous hands (Li et al. 2025), and humanoid robots (Cao 2025). As planning fundamentally defines the high-level decision layer guiding low-level control, enabling reliable autonomous planning is a cornerstone of embodied AI (Li et al. 2025). Existing surveys have provided detailed categorization and analysis for different aspects of autonomous planning, including task planning and motion planning (Guo et al. 2023; Claussmann et al. 2019; Schwarting et al. 2018).

To achieve higher efficiency and scalability in real-world deployments, multi-agent systems have become increasingly important. Multiple autonomous agents collaborating in shared environments can significantly improve performance, coverage, robustness, and operational flexibility (An et al. 2023). However, developing cooperative multi-agent autonomous planning is extremely challenging. Such systems must maintain coordination and efficiency as the number of agents increases, avoid collisions with both dynamic and static obstacles, operate under limited or unreliable communication bandwidth, and handle uncertainties from sensing, localization and environment dynamics (Gielis et al. 2022; Prorok et al. 2021). These challenges make multi-agent planning substantially more complex than single-agent planning.

Although TAMP has been extensively studied across autonomous agents, existing surveys mainly focus on either single-agent systems, or separately review task allocation, multi-agent path finding, or motion planning in isolation. However, TAMP in multi-agent systems fundamentally couples these problems. This survey aims to systematically break down this complexity and provide a comprehensive understanding of multi-agent Task and Motion Planning (TAMP). The remainder of this article is organized as follows: Sects. 2.1, 2.2 and 2.3 provide background overviews of TAMP, MATA and MAPF. Section 3 introduces a new taxonomy of Multi-agent TAMP solution strategies, and quantifies papers published in the past five years under this classification. Sections 4 and 5 summarize the recent trends and future research directions in multi-agent TAMP. Finally, Sect. 6 concludes this survey.

2 Background

2.1 Task and motion planning (TAMP)

TAMP is also called hybrid planning, which is essentially integrating high-level, discrete actions with corresponding low-level, continuous paths to achieve a goal state (Dantam 2020; Mansouri et al. 2021). In past traditional agent planning research, task planning and motion planning are usually developed separately. But with the planning problems becoming more complex, more and more researchers found that considering motion planning can help better improve task planning efficiency (Mansouri et al. 2021; Antonyshyn et al. 2023). First, under resource limited situation or with geometric or kinematic constraints, TAMP can generate feasible plans more quickly with less cost in computing (Zhang and Shah 2016). Second, combining motion planning with task planning can help avoid, eliminate, or mitigate inter-agent collisions (Kim et al. 2019). Third, in real-time or autonomous scenarios, task and motion planning are required to be determined simultaneously (Jiang et al. 2019).

Garrett et al. (2021) published a detailed review on how to solve TAMP problems. In brief, task planning is formulated as a discrete planning problem and addressed using symbolic reasoning. Motion planning, more precisely multi-modal motion planning (MMMP), is regarded as a hybrid constraint satisfaction problem (H-CSP), which involves searching for both discrete and continuous parameter solutions. Solving a integrated TAMP problem can be viewed as constructing a plan skeleton with specific parameter values. In particular, it involves selecting an appropriate action sequence and assigning the necessary parameters to each action by solving the H-CSP.

As to the order of action sequencing and H-CSP solving, Garrett et al. (2021) concluded into three main strategies: sequencing first, which finds whole plan skeletons and then check their geometric and kinematic accessibility to satisfy all the constraints; satisfying first, which sampling sets of satisfying parameter values for individual constraints and then assemble actions using those values into complete plans; and interleaved, in which actions added to the plan and constraints are satisfied incrementally. The first two strategies have been widely used, the core idea of both is to separate task and motion planning processes and make them a loop (Savla et al. 2005; Dantam 2020). When considering multi-agent systems, naively performing task and motion planning sequentially is usually infeasible (Antonyshyn et al. 2023). Therefore, this survey aims to provide an overview of various multi-agent TAMP solution strategies that have been proposed in recent research.

Currently, the mainstream research trends of TAMP include long-horizon planning (Hartmann et al. 2023), belief-space planning (Zhao et al. 2025), learning-based planning (Dong et al. 2023), and multi-agent planning (Wang et al. 2025). This survey will focus on the extensions of multi-agent planning to the TAMP domain.

2.2 Multi-agent task allocation (MATA)

Task planning is a process of generating a sequence of discrete actions which aims to achieve a target goal state (Gayathri and Uma 2018). In multi-agent systems, task planning usually involves three sub-problems: task decomposition, task allocation and task scheduling. Task decomposition is a process of sub-dividing a big complex or compound task into a series of simple small tasks which are easy to perform. Task scheduling is a process of ordering tasks

in an individual robot. These two problems are both common in multi and single robot systems. While task allocation, which means assigning tasks to different agents to achieve the goal state, is the core challenge in multi-agents systems. Many formulations of the MATA problem have been shown to be NP-hard, particularly when considering dynamic environments, temporal constraints, or large-scale agent-task interactions. Dynamic task allocation algorithms can be centralized or decentralized, depending on the communication topology, and can involve homogeneous or heterogeneous agents. The main strategies used for solving MATA include auction based algorithms (Braquet and Bakolas 2021; Alshaboti and Baroudi 2021; Schneider et al. 2015), optimization based algorithms (Patsias et al. 2023; Huang et al. 2021; Wang et al. 2017), game theory based algorithms (Martin et al. 2023; Rahman et al. 2019; Zhang et al. 2018) and learning based algorithms (Ma et al. 2024; Liu et al. 2022; Ji et al. 2021; Wei et al. 2020). In some research, not only one kind of these algorithms is used. Li et al. (2023) formulated apple harvesting problems through multi-arm as a multi-arm task planning process based on the Markov game, and solved it through reinforcement learning. At present, learning based algorithms are a hot topic.

In auction based algorithms, tasks are allocated through negotiation protocols and bidding based on the utility or calculated cost, and the allocation process is usually centralized and relying on a central agent playing the role of an auctioneer (Smith 1980; Khamis et al. 2015). But it does not mean that the auction-based algorithms can not be done in a decentralized way, that might take several rounds. The consensus based bundle algorithm and the contract net protocol are two representative auction based algorithms. Bai et al. (2022) proposed a group-based distributed auction algorithm for MATA, which is a two-stage distributed optimization framework, inspired by ridesharing allocation algorithms.

The optimization based algorithms are a mathematics progress aiming to find the optimal solutions based on their cost functions which depend on profits, objectives, constraints and so on. The algorithms can be either deterministic or stochastic (Skaltsis et al. 2021). Deterministic algorithms include graph based algorithms (Xu and Song 2023; Wang et al. 2022), mixed integer linear programming (Lippi and Marino 2021; Afonso et al. 2020), etc. All these algorithms follow reproducible computational paths as long as the starting point is the same, and are therefore referred to as deterministic algorithms. Stochastic algorithms are also called metaheuristics algorithms, which include several algorithms like swarm intelligence (Yasser et al. 2024; Schwarzrock et al. 2018), genetic algorithms (Martin et al. 2021; Rekha and Dakshayini 2019; Page et al. 2010), simulated annealing (Mahjoubi et al. 2024; Wang et al. 2022) and others. Choudhury et al. (2022) proposed a hierarchical task allocation algorithm, Stochastic Conflict-Based Allocation, which uses multi-agent conflict resolution with single-agent policy tree search. In particular, the low level independently determines the optimal task attempt sequence for each individual agent, ignoring other agents; the high level resolves potential conflicts in assigned tasks across multiple agents to obtain a valid multi-robot allocation.

The game theory based algorithms mean that agents act like players and get payoff at the end of the game. The strategies taken will decide their action and lead to different payoff. So usually when the agents have chosen the best strategy, they tend to stick on it to keep the best outcome, and finally they will reach a condition called Nash equilibrium. The game can be either cooperative or competitive. Depending on this, agents will show difference on their action response, forming coalitions or being selfish to reach the highest payoff. Zhao et al. (2021) formulated the fairness-aware task assignment problem in spatial crowdsourc-

ing as a multi-player game. Under this setting, they developed a Fairness-aware Game-Theoretic algorithm to sequentially and asynchronously update workers' strategies until achieved a satisfying task allocation when a pure Nash equilibrium is reached.

Learning based algorithms have become a major focus of current research. Such algorithms enable MATA problems to be solved in dynamic environments and have found wide application in various specialized domains beyond domestic settings. The core theory of learning based algorithms is forming the environment into a Markov Decision Process (MDP) and enabling agents to make use of past solutions and other agents' solutions to generate their own new solution (Tanaka and Yamamura 2003). A typical method is reinforcement learning (RL) (Zhao et al. 2023; Zhang et al. 2022; Yin et al. 2022; Lee et al. 2022). RL has gained widespread adoption owing to its capability to handle environmental uncertainties, support real-time implementation for well-trained networks, and adapt to diverse tasks (Skaltsis et al. 2021). Dai et al. (2025) utilized RL to obtain a decentralized policy for each agent to realize task allocation and scheduling iteratively. Chen et al. (2022) formulated the long-term dynamic task allocation and service migration problem in edge-cloud IoT as a MDP and proposed a deep RL approach to solve it. Q-learning is a representative and most widely used algorithm of RL (Ji et al. 2021). To tackle task allocation challenge in mobile crowd-sensing, Akter et al. (2021) utilized a deep Q-learning based algorithm while Ji et al. (2023) proposed a new high-level Q-learning based hyperheuristic evolutionary algorithm which is designed to find a suitable low-level heuristic for each state.

There are two main communication schemes in MATA, the centralized (Liu and Kroll 2012; Chakraa et al. 2023; Msala et al. 2023) and the decentralized (Wang et al. 2023; Yue et al. 2025; Flushing et al. 2016). Most algorithms used the central control communication unit (Hwang et al. 2022; Wang et al. 2022; Zhang et al. 2025; Yan and Di 2023). The advantages are that they require fewer system resources. But they require more computational resources and the feasibility is only guaranteed in a small amount of agents (Zitouni et al. 2020; Skaltsis et al. 2021). The decentralized systems allow the agents to negotiate with each other instead of a central agent. In particular, every agent has its own utility function and has a local perception of the environment. As a result, the decision for MATA is made locally and distributively. Hamidoglu et al. (2025) proposed a cost-efficient Nash-based game framework, which relies on a decentralized decision-making strategy, for task allocation in a distributed robotic network. The advantages of decentralized systems are that they are robust and more scalable with less computational cost compared to centralized systems. But they require additional consensus algorithm and their solutions of MATA are usually suboptimal.

A comprehensive discussion on the advantages and disadvantages of different algorithms, as well as their communication mechanisms, can be found in the survey by Skaltsis et al. (2021).

2.3 Multi-agent path finding (MAPF)

Motion planning involves finding a feasible path or trajectory within the agent's state space, ensuring the transition from a start state to a goal state respects both geometric and dynamic constraints. In multi-agent systems, the major challenge of motion planning lies in the MAPF problem. The goal of MAPF is to find a non-colliding path for each agent while minimizing their travel times.

The MAPF problem can be viewed as a generalization of the single-agent path planning problem. Its algorithms are conceptually similar, typically following two main paradigms: centralized and decentralized. Centralized algorithms typically include search-based algorithms such as A* (Garcia et al. 2023; Wang and Goh 2011), increasing cost fee search (ICTS) (Ribeiro et al. 2025; Wang et al. 2024), Conflict-Based Search (CBS) (Moldagalieva et al. 2024; Qiao et al. 2022; Kottinger et al. 2022). In addition, sampling-based methods such as Probabilistic roadmap (PRM) (Jathunga and Rajapaksha 2025; Kumar and Sikander 2024) and Rapidly-exploring Random Tree Star (RRT*) (McBeth et al. 2023; Cain et al. 2021). Building upon these foundational algorithms, several variants have been developed. For instance, Theta* extends A* algorithm to enable any-angle path planning on grids (Daniel et al. 2010). Binary Branching Multi-Objective Conflict-Based Search (BB-MO-CBS) extends CBS to handle multi-objective optimization criteria (Ren et al. 2023). Centralized paradigms generally guarantee global optimality and enable complex cooperative behaviors, but they tend to be computationally expensive, hard to scale, and require global communication and synchronization among agents.

Decentralized algorithms, in contrast, rely on distributed decision-making and local information exchange among agents. Representative reactive approaches include Reciprocal Velocity Obstacle (RVO) (Chen et al. 2024; Han et al. 2022) and Optimal Reciprocal Collision Avoidance (ORCA) (Ruiz-Fernandez et al. 2025; Dai et al. 2025), which evolved from the concept of Velocity Obstacle (VO). Unlike centralized algorithms that pre-calculate entire paths, these reactive-based algorithms enable agents to navigate safely by making local adjustments based on the relative velocities of neighbors. These algorithms enable reciprocal collision avoidance through geometric reasoning in velocity space and are widely adopted for real-time multi-robot path planning. Beyond purely reactive schemes, decentralized strategies also encompass Model Predictive Control (MPC) (Khan et al. 2025; Xin et al. 2024), Artificial Potential Fields (APF) (Wang et al. 2024; Shao et al. 2023), and Consensus-Based Approaches (Wozniak 2025; Yildiz et al. 2024) etc. These algorithms are typically scalable to large robot swarms, robust to failures, and suitable for real-time applications. However, they cannot easily guarantee global optimality and may suffer from deadlocks or local minima.

There are also other algorithms, including rule-based algorithms such as Push and Swap (Alotaibi and Al-Rawi 2018; Luna and Bekris 2011), push and rotate (De Wilde et al. 2014). Decoupled algorithms, including prioritized planning (Binder et al. 2019; Zhang et al. 2022), decompose the multi-agent planning problem into sequential single-agent subproblems based on priority assignments. Optimization-based extensions have also emerged to enhance the performance of classical MAPF solvers. For example, Li et al. (2022) proposed MAPF-LNS2, a Large Neighborhood Search based algorithm, to solve MAPF. Zang et al. (2025) improved the solution quality of a rule-based MAPF algorithm through optimizing a policy to generate adaptive guidance. Yan et al. (2025) proposed kinodynamic Temporal Plan Graph Planning (kTPG), a multi-agent velocity optimization algorithm, to refine a MAPF plan into a kino-dynamically feasible plan. Zhang et al. (2024) proposed two Guidance Graph Optimization (GGO) algorithms to automatically generate high-quality guidance information, significantly improving the throughput of lifelong MAPF systems.

Recent studies indicate that standalone MAPF solvers and their incremental variants are increasingly insufficient for addressing large-scale, dynamic, and real-world MAPF scenarios. With the rapid growth of computational resources and advances in machine learning,

hybrid (Garcia et al. 2023; Almadhoun et al. 2022; Das et al. 2016) and learning-based (Cui et al. 2024; Zhu et al. 2021; Wen et al. 2021; Wang et al. 2021; He et al. 2020) approaches have recently gained increasing attention. Learning-based representative examples include Graph Neural Network (GNN) planners (Li et al. 2021), which learn decentralized policies that mimic centralized solutions; Reinforcement Learning (RL) (Wen et al. 2021; Chen et al. 2023), which trains agents to coordinate implicitly; recurrent neural networks (RNN) (Zhu et al. 2021), which are usually designed to address sequential problems. Hybrid approaches not only combine multiple classical MAPF algorithms, but also integrate traditional planning algorithms with learning or optimization modules to balance structure and adaptability. For instance, Tang et al. (Tang et al. 2025) developed a centralized learning-based policy for MAPF problem by collecting trajectories from rule-based methods and training their model in a supervised way. Their policy can generalize across different maps and handle any number of agents by leveraging a CNN-based architecture. Tong et al. (2022) combined A* and improved APF. Wang et al. (2024) address the Multi-Objective MAPF problem by leveraging the single-agent multi-objective search algorithm A*pex to accelerate different components of the BB-MO-CBS framework. Dergachev et al. (2020) integrated three algorithmic components (Theta*, ORCA and Push and Rotate) to solve MAPF problems. Despite their promising performance, hybrid and learning-based approaches often incur substantial computational overhead and require extensive training data, which may limit their practicality in resource-constrained or safety-critical deployments. As highlighted in the survey by Jiang et al. (2024), a key future direction lies in bridging the gap between idealized MAPF models commonly used in academic benchmarks and the constraints of real-world robotic systems.

3 Taxonomy of multi-agent TAMP solutions

Existing taxonomies have mostly focused on TAMP problems in single-robot systems (Garrett et al. 2021) or on task allocation (Ka and Subramaniam 2024) or path planning (Madrignano et al. 2021) separately in multi-robot contexts. Antonyshyn et al. (2023) proposed a taxonomy for multi-agent TAMP problems, but their taxonomy primarily focuses on problem formulation and thus provides limited guidance on how to solve TAMP problems. Guo et al. (2023) conducted a survey on recent trends in multi-robot TAMP but emphasized online and learning-based algorithms. To address this gap, we propose a comprehensive and systematic taxonomy of solution strategies for TAMP in multi-agent systems. The proposed taxonomy evaluates multi-agent TAMP solutions along three key dimensions: integrated structures, communication systems and planner adaptability. The corresponding criteria for each dimension are outlined below.

3.1 Integrated structures: task planning first, motion planning first, interleaved planning

According to the taxonomy of Garrett et al. (2021) used for describing how to classify single-agent TAMP algorithms, the multi-agent TAMP problem solutions can be also divided into three categories:

Task planning first: Generates action sequence first and then finds the continuous parameter values satisfying constraints. This procedure can also be iteratively repeated.

Motion planning first: Generates a series of continuous parameter values satisfying constraints first, and then choose available actions and sequence them. This procedure can also be iteratively repeated.

Interleaved planning: Whenever either the action-sequence generation or the parameter-constraint satisfaction process is partially completed, the planner switches to the other process, interleaving the two to progressively construct the complete plan.

3.2 Communication systems: centralized and decentralized

According to the communication system, TAMP can be separated into centralized and decentralized types. The most common way is relying on central processors to compute centrally and make decisions, yet it offers a high requirement on computing and telecommunication capability. The decentralized system shows its advantage in the flexibility and computing power saving. It enables each agent to make its own decisions in a distributed manner, but it's hard to work when the structure becomes too complex (Chen et al. 2021; Jia et al. 2024).

Centralized system: In this kind of system, a central processor is responsible for computing and decision-making, ensuring coordinated control.

Decentralized system: Individual units or agents make their own decisions in a distributed manner based on local information instead of obeying a central controller, enhancing flexibility and reducing computational load.

During classification, we found that some research adopt hybrid hierarchical architectures that combine both centralized and decentralized communication mechanisms. However, since centralized systems represent the traditional and baseline communication paradigm, and to ensure a clear and concise taxonomy framework, our goal is to highlight the fundamental differences in communication structures rather than introduce overly fine-grained distinctions. Therefore, for consistency and clarity, any framework that incorporates decentralized communication components is categorized under the decentralized class in our taxonomy. Furthermore, to avoid potential confusion between the terms distributed and decentralized, we explicitly clarify their distinction in this survey. The term distributed refers to the physical arrangement of computation, indicating that computational processes are executed across multiple agents or nodes. Decentralized refers to the authority structure of decision-making, where no single agent serves as a central controller, and each agent makes decisions autonomously based on locally available information and limited coordination with others.

3.3 Planner adaptability: fixed and adaptive

When an unexpected situation occurs, there are generally three possible ways to address it. The first and most straightforward approach is to cancel the current plan, return to the home point, and then attempt to generate a new plan. This method is very safe but lacks flexibility and is time-consuming, particularly in emergency situations. The second approach is to revert to the previous state and attempt the task again. However, this only works when the unexpected situation is caused by temporary or accidental factors, such as a sudden gust of wind that blows the drone off its intended path.

The third approach is to generate a new plan, commonly referred to as adaptive planning. From a task-level perspective, the new plan can either be a fixed task plan with an updated trajectory or path or an entirely new task plan with its corresponding trajectory or path. However, a well-defined criterion is required to determine when a task plan should be modified. This process can be regarded as a form of probabilistic planning. There are generally two types of solutions for generating new plans: one that makes use of previous plans, and another that relies on situation awareness to generate a completely new one. Both solutions can benefit from feedback controllers and reactive techniques; however, the latter imposes higher requirements. It depends on real-time communication between system modules and strong computational capability. At present, this has become an active area of research, often associated with online planning and edge computing. In this scale, we consider the first and second way mentioned above as fixed type, and the third way, generating a new plan, as an adaptive type.

Fixed plan: A problem-solving approach in which a predetermined plan is followed. Adjustments typically involve reverting to a previous state or canceling the plan entirely.

Adaptive plan: A flexible problem-solving approach that dynamically generates partial or essential new plans in response to environmental changes, real-time feedback, or inherent uncertainties after or during executing. It can adjust high-level discrete task sequences and low-level continuous motion trajectories.

3.4 Classification and analysis

The proposed taxonomy consists of three conceptually distinct and orthogonal dimensions, each capturing an independent aspect of the solution design. Integrated structures describe the coordination mechanism between task and motion planning, communication systems define how information exchange and decision-making are organized across agents, and planner adaptability characterizes whether planning is performed offline or dynamically updated during execution. This separation enables systematic classification of multi-agent TAMP solutions without imposing structural constraints.

The proposed taxonomy categorizes and analyzes relevant studies published in the past five years (2021–2025), revealing the major research trends and popular approaches for solving multi-agent TAMP problems in Table 1 and Table 2. To ensure the comprehensiveness, representativeness, and reproducibility of the literature review, a systematic literature screening process was conducted using multiple major academic databases, including IEEE Xplore, Elsevier ScienceDirect, SpringerLink, ACM Digital Library, Wiley Online Library, and Google Scholar. These databases were selected because they collectively cover the most influential journals and conference proceedings in robotics, artificial intelligence, and multi-agent systems.

The literature search was performed using combinations of the following keywords: “TAMP”, “multi-agent systems” and “task and path planning”, as well as related variations and combinations of these terms. The inclusion criteria required that the studies explicitly address multi-agent TAMP problems, where agents may consist of multiple robots or human–robot collaborative systems, and the TAMP process must involve integrated task planning and path planning components. Literature that focuses exclusively on sub-problems of TAMP, such as MATA or MAPF, without considering the complete task–motion integration process, was excluded. Additionally, selected literature was chosen to ensure

Table 1 Fixed plan

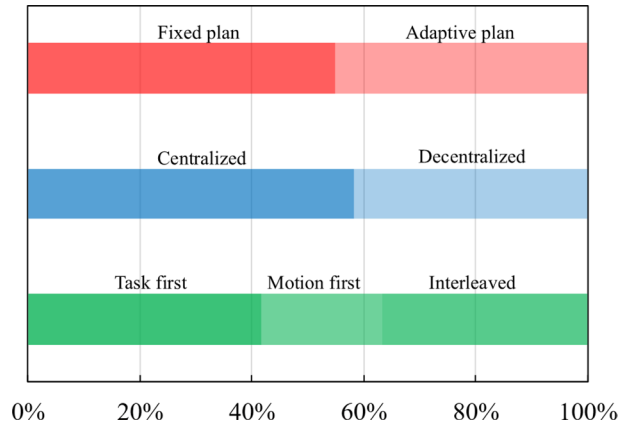
	Fixed plan		
	Task first	Motion first	Interleaved
Centralized	(Zeng and Li 2026; Wang et al. 2026, 2025; Yu et al. 2025; Wang et al. 2023; Rajendran and O'brien 2022; Yan et al. 2021; Xia et al. 2021; Liu et al. 2021; Touzani et al. 2021)	(Zhang et al. 2024, 2023; Motes et al. 2023; Li and Ma 2023; Alitappeh and Jed-disaravi 2022; Gundana and Kress-Gazit 2021)	(Li et al. 2026; Xidias and Zacharia 2024; Xu et al. 2024; Lee and Long 2023; Ren et al. 2023; Luo and Zavlanos 2022; Zhang et al. 2022; Chen et al. 2021; Salvado et al. 2021)
Decentralized	(Mandi et al. 2024; Zhang et al. 2024; Bai et al. 2022)	(Okubo and Takahashi 2023; Lindemann et al. 2021)	(Fang et al. 2024; Lei et al. 2023; Bai et al. 2022)

Table 2 Adaptive Plan

	Adaptive plan		
	Task first	Motion first	Interleaved
Centralized	(Liu et al. 2025; Faroni et al. 2024; Zhang et al. 2024; Galati et al. 2023; Bolu and Korçak 2021)	(Li et al. 2026; Yamauchi et al. 2022; Salvado et al. 2021)	(Liu et al. 2022; Choudhury et al. 2022)
Decentralized	(Yang et al. 2026; Xu et al. 2024; Zhao et al. 2024; Feng et al. 2024; Peti et al. 2023; Karami et al. 2021; Chen et al. 2021)	(Newaz et al. 2023; Contini and Farinelli 2021)	(David and Valencia 2025; Li et al. 2024; Peng et al. 2024; Chen et al. 2024; Xu et al. 2023; Zhu et al. 2023; Agrawal et al. 2022; Liu et al. 2021)

broad coverage across different taxonomy dimensions proposed in this work. This selection strategy ensures that the reviewed literature provides a representative and balanced overview of the state-of-the-art (SOTA) methods in multi-agent TAMP.

In total, 60 papers were included in the taxonomy. As shown in Fig. 1, from the perspective of integrated structures, 41.67% of the papers adopt a task-first architecture, 21.67% adopt a motion-first architecture, and 36.66% employ an interleaved structure. In terms of communication systems, 58.33% of the works follow a centralized approach, while 41.67% adopt a decentralized one. Regarding planner adaptability, 55.00% of the planners are fixed, whereas 45.00% are adaptive. The most prevalent solution strategy for multi-agent TAMP is the task first, centralized system with a fixed plan, which accounts for 16.67% of the

Fig. 1 Analysis of included papers from 2021-2025

reviewed papers. The second most common approach is the interleaved, centralized system with a fixed plan, representing 15.00% of the total.

The dominance of task-first, centralized, fixed planning frameworks can be largely attributed to their technical convenience and implementation tractability. The task-first decomposition simplifies the overall TAMP problem by separating task allocation from motion planning, allowing researchers to leverage mature combinatorial optimization methods without explicitly addressing tightly coupled task–motion dependencies. Centralized architectures further facilitate this process by providing global system information, enabling straightforward coordination and often improving solution optimality under complete knowledge assumptions. In addition, fixed planning avoids the computational and algorithmic complexity associated with real-time re-planning, making such frameworks easier to implement and evaluate, particularly in static or structured environments commonly used for algorithm validation. A comparison between the most prevalent (task-first, centralized, fixed planning frameworks) and second most common strategies (interleaved, centralized, fixed planning frameworks) further reveals that the primary variation lies in the integrated structure, while the communication system and planner adaptability remain unchanged. This pattern can be attributed to the different levels of implementation complexity associated with each dimension. Modifying the integrated structure primarily involves algorithmic redesign of the task–motion coordination process, which can be achieved without fundamentally altering the system architecture. From a design perspective, interleaved strategies can be viewed as a transitional paradigm that incrementally increases task–motion coupling without incurring the full complexity of adaptive and decentralized systems. In contrast, transitioning from centralized to decentralized communication requires significant changes to information exchange mechanisms and coordination protocols, while introducing adaptive planning imposes substantial computational and real-time decision-making requirements. As a result, researchers often explore interleaved structures as an intermediate step to improve task–motion coupling and solution quality while retaining the technical simplicity and stability of centralized and fixed planning frameworks.

Moreover, functional coupling often emerges between these dimensions due to system-level requirements such as scalability, computational efficiency, and responsiveness to environmental changes. For example, adaptive planning requires frequent re-planning based on updated information, which introduces substantial computational and communication

demands. Decentralized communication systems naturally support such adaptability by distributing computation and enabling local decision-making, thereby improving responsiveness and robustness. This tendency is also reflected in the conditional heatmap (Fig. 2), where approximately 63.00% of adaptive planning approaches adopt decentralized communication systems. In contrast, an opposite coupling pattern can be observed for fixed planning strategies. Among centralized approaches, fixed plan accounts for approximately 71.00%. These performances suggest that these design choices are not independent, but rather co-evolve due to shared computational and communication constraints. Overall, the observed design patterns reflect an inherent trade-off between solution quality, computational complexity, and system scalability, which continues to shape the evolution of multi-agent TAMP frameworks.

4 Recent five years trends

The taxonomy of the planning problem follows the classification proposed by Antonyshyn et al. for multi-robot TAMP (Antonyshyn et al. 2023), including combinatorial optimization (CO), symbolic representations (SR), Markov decision process (MDP), graph-based approaches (GBA) and game theory (GT).

4.1 Planner structure design

In recent years, the design of planner structures for TAMP has become a popular research topic. The most common structures include hierarchical and interleaved architectures. The interleaved structure typically relies on optimization algorithms to iteratively obtain the

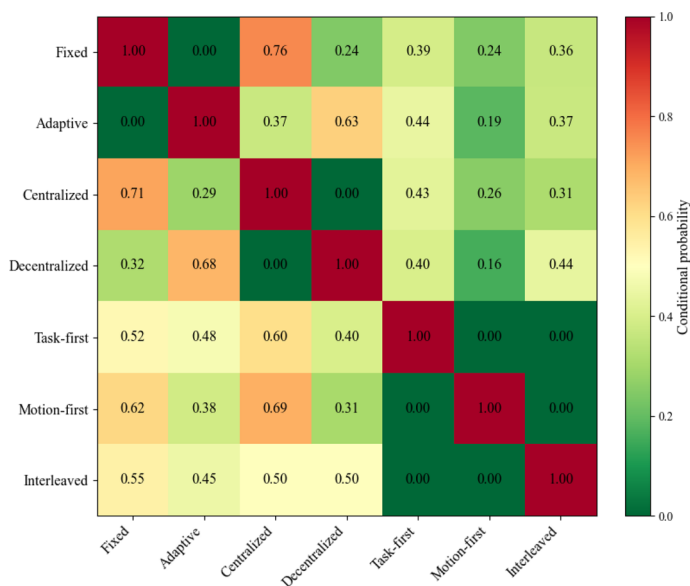


Fig. 2 Conditional heatmap of the relationships among taxonomy dimension. Each cell represents the conditional probability $P(\text{column} \mid \text{row})$

optimal solution, where the outputs of the task planner and motion planner serve as optimization objectives for each other within the optimization process. Due to its ability to decouple complex TAMP problems into subproblems, the hierarchical structure greatly simplifies the planning process and is well-suited for addressing online planning problems. It has therefore gained significant attention in recent years. Table 3 summarizes representative papers corresponding to the structures discussed above.

Chen et al. (2021) propose a new approach to address multi-agent pickup and delivery (MAPD) problem. The authors break away from the traditional MAPD solution paradigm, which first allocates tasks and then plans paths. Instead, their approach integrates path coordination costs directly into the task allocation process. Specifically, they replace the conventional task allocation metrics, typically based on estimated distances or shortest-path costs, with marginal costs. To evaluate the true coordination cost of different task allocations, the approach employs prioritized planning as a fast path planning evaluator. Finally, Large Neighborhood Search (LNS), a meta-heuristic optimization strategy, is applied to iteratively refine the solution based on the current allocation. Overall, the approach follows an assignment–coordination–refinement cycle.

Fang et al. (2024) proposed a three layer planner for TAMP which had been proved effective through experiments. In the upper layer, task allocation layer, a standard Genetic Algorithm (GA) is adopted to iterate and evolve based on the path lengths and the probability of conflicts, and assigns tasks to agents according to scheduling order. Secondly, an Improved Car-Like Conflict-Based Search (ICL-CBS) algorithm is presented as the conflict resolution layer to reduce the runtime. At the bottom, path planning layer, ICL-CBS and the

Table 3 Planner Structure design

References	(Chen et al. 2021)	(Fang et al. 2024)	(Faroni et al. 2024)	(Liu et al. 2021)
Planning problem	CO	CO	SR	GBA
Task planning algorithms	Heuristic and LNS	GA	PDDL and heuristic	Probabilistic Traffic Model and GCG and Greedy Near-Optimal
Motion planning algorithms	Prioritized Planning	ICL-CBS and SHA*	Sample based	CBS
integrated structure	Interleaved	Interleaved	Task first	Interleaved
Communication structure	Centralized	Centralized	Centralized	Decentralized
Planner adaptability	Fixed	Fixed	Adaptive	Adaptive
Planner structure	Interleaved: assignment-coordinationrefinement loop	Hierarchical: three layers (task allocation, conflict resolution, path planning)	Hierarchical: two layers (task planner, action)	Hierarchical: two layers (integrated task allocation and sector-Level path planning, low-level path coordination)

LNS: Large Neighborhood Search; GA: Genetic Algorithm; ICL-CBS: Improved Car-Like Conflict-Based Search; SHA*: Spatiotemporal Hybrid-State A*; GCG: Generalized Conflict-Graph

Spatiotemporal Hybrid-State A* (SHA*) algorithm are jointly used to meet the kinematic constraints and find the path according to the task sequences.

Faroni et al. (2024) proposed a hierarchical planner where the high layer, task planner, is responsible for symbolic task planning and optimization under temporal constraints. The low layer, action layer, turns the symbolic task into feasible actions in time and selects the optimal actions to generate motion plans online. The generated motion plans will be sent to motion controller to conduct. If the action layer or motion controller fails, the task planning layer will re-plan, following the Sense–Plan–Act paradigm. The proposed structure can effectively handle task planning problems with temporal constraints and variable task durations, while enabling online motion planning to cope with environmental changes.

Liu et al. (2021) proposed a hierarchical planning and coordination framework, which consists of two layers: a high-level centralized planning module and a low-level distributed coordination module. The high-level module, integrated task allocation and sector-level path planning, operates on a sector topology graph. It assigns tasks and generates macro-level routes across sectors for each robot through three submodules: candidate robot task and path generation, motion conflict prediction, and generalized conflict-graph (GCG) based optimization. The low-level module, path coordination, operates on the local road topology graph and runs independently within each sector as a distributed execution layer. It adopts a rolling horizon mechanism to dynamically generate real-time, fine-grained road-level paths. This framework has been demonstrated to improve the efficiency, robustness, and scalability of large-scale multi-robot systems.

4.2 LLM based TAMP

Large Language Models (LLMs), owing to their strong reasoning and decision-making capabilities, have recently been widely explored for single-agent task planning (Wang et al. 2025; Ao et al. 2025; Song et al. 2024; Chen et al. 2024). Existing research primarily focuses on leveraging LLMs to help planners interpret abstract semantic goals and compensate for the lack of commonsense knowledge during planning (Yang et al. 2025; Izquierdo-Badiola et al. 2024). This is typically achieved through carefully designed initial prompts (e.g., in-context learning, chain-of-thought) or iterative re-prompting mechanisms that incorporate feedback, such as environment state updates or detected execution errors (Sun et al. 2025; Yang et al. 2025). In addition, some research attempts to directly train or prompt LLMs to generate executable plans (Lin et al. 2023; Joubin et al. 2024). Given these successes, there is new interest in investigating LLMs as task or TAMP planners for multi-agent systems. Table 4 summarizes the corresponding representative papers.

Mandi et al. (2024) proposed RoCo, a unified approach for multi-robot TAMP that leverages LLMs for both high-level task coordination and low-level motion planning. RoCo is composed of three parts: dialogue-style task coordination, feedback-improved sub-task plans generated by LLMs and LLM-informed motion planning in joint space. From the perspective of functional roles, LLMs contribute differently across the task and motion planning layers. At the task planning level, LLMs replace traditional symbolic planners by generating sub-goals through iterative dialogue and contextual reasoning, enabling flexible decomposition of high-level objectives. At the motion planning level, LLMs act as high-level prompters, providing RRT with waypoint paths that incorporate high-level task semantics and environmental constraints, thereby reducing the motion planner's sample

Table 4 LLM based TAMP

References	(Mandi et al. 2024)	(Chen et al. 2024)	(Wang et al. 2025)	(Zhang et al. 2024)	(Liu et al. 2025)
Planning problem	SR	SR	SR	SR	SR
Task planning algorithms	LLM	LLM	LLM	LLM	LLM
Motion planning algorithms	LLM and RRT	LLM	LLM	VLM ¹ and Multi-Level Dijkstra and A*	A*
integrated structure	Task first	Task first	Task first	Task first	Task first
Communication structure	Decentralized	Centralized	Hybrid ²	Centralized	Decentralized
Planner adaptability	Fixed	Adaptive	Fixed	Fixed	Adaptive
LLMs function in task planning	Generate sub-goals	Decompose and allocate tasks	Generate actions	Generate sub-task sequences	Generate action list
LLMs function in motion planning	Generate way-point paths	Validate actions and perform self-reflection	Validate and complete actions with feedbacks	None	None

¹VLM: Visual Language Model

²Hybrid means there are not only one kinds of communication systems used in this reference

complexity. Similar efforts have leveraged feedback-driven prompting to enhance the planning capability of LLMs. For instance, Liu et al. (2025) proposed an LLM-based centralized hierarchical framework for heterogeneous multi-robot systems. The framework conducts multiple Proposal-Execution-Feedback-Adjustment cycles between a centralized task allocator and individual robot executors to finish a complex long-horizon task. Particularly, each robot is equipped with an independent LLM and gives low-level execution feedback to the centralized task allocator, enabling bottom-up task correction in the next iteration.

For LLM-based multi-agent TAMP systems, a common strategy is to assign an independent LLM instance to each agent to enable decentralized reasoning and decision-making. However, as the number of agents increases, this approach introduces several scalability challenges. Specifically, the cumulative context length grows rapidly, approaching the token limits of current LLMs, while also increasing inference latency and API costs. Moreover, relevant information may become diluted within lengthy prompts, reducing reasoning efficiency and decision quality. To address these challenges, Chen et al. (2024) compared the task success rate and token efficiency of four communication systems (centralized, decentralized, and two hybrid) in LLM-based planners for multiple robots cooperation, and found that a hybrid system achieved outstanding performance. Their research reveals that a single centralized LLM planner tends to generate suboptimal or infeasible plans when dealing with complex multi-agent coordination scenarios. Conversely, decentralized multi-LLM systems typically require multiple rounds of inter-agent dialogue at each planning step, leading to prolonged decision latency. During such interactions, agents may redundantly repeat previously shared information or repeatedly propose identical actions without contributing novel insights, resulting in inefficient communication and contextual dilution. However, the hybrid communication structure represented by HMAS-2 can balance coordination effectiveness and computational efficiency. In this framework, each robot is equipped with a

local LLM responsible for generating agent-specific observations and feedback, which are then communicated to a central LLM planner responsible for global coordination and task-level decision-making. Their work provides important insights into the role of communication architecture in determining the effectiveness and scalability of LLM-based multi-agent TAMP systems, and helps bridge a critical gap in understanding how LLM coordination mechanisms influence multi-agent planning performance.

Wang et al. (2025) proposed a new system which utilized an LLM for task planning to address the last-mile delivery problem. The LLM-based task planning module contains three phases. The first one is responsible for extracting and parsing a series of addresses from a natural language delivery request, and subdivides them into hierarchical sub-addresses. The second one is employed to verify the extracted information to mitigate LLM's hallucinations. The last one is responsible for grouping geographically proximate tasks and optimizing the task sequence. In essence, the LLM serves as a semantic interpretation module responsible for task decomposition and scheduling, but does not directly participate in low-level path generation. Compared to other LLM-based planning approaches, this work explicitly addresses the hallucination problem, a critical reliability concern in embodied AI. As reported by Ravichander et al. (2025) and Liu et al. (2026), even SOTA models such as GPT-5 and Gemini-2.5-Pro suffer from severe hallucination issues, with up to 86% of generated atomic facts being incorrect and step localization accuracy as low as 41.1%. To mitigate this, the authors introduce a secondary prompt mechanism to verify extracted information and guide the LLM toward more human-like reasoning during validation, thereby reducing the risk of semantic errors propagating into downstream planning modules and improving overall system robustness.

Zhang et al. (2024) proposed a cognitive-inspired modular framework to realized multiple embodied agents cooperation. In this framework, each agent runs an independent LLM and coordinates with others through dialogue. The LLM simultaneously functions as both a task planner and a communication channel, enabling the agents to establish a shared symbolic space via natural language interaction. Importantly, the authors introduce a dedicated execution module instead of relying on the LLM to directly generate primitive control actions. This design choice is motivated by the observation that, while LLMs are effective at generating high-level plans and reasoning over abstract tasks, they exhibit limited reliability and precision in low-level control generation. By decoupling high-level planning from low-level execution, the framework preserves the semantic reasoning strengths of LLMs while ensuring robust and physically feasible action execution through conventional control modules.

Based on the above analysis, prior research has extensively explored the planning capabilities of LLMs and established several emerging consensus findings. In particular, LLMs exhibit strong high-level reasoning and decision-making capabilities, enabling effective interpretation of semantic goals and decomposition of complex tasks into structured sub-goals. However, they remain less reliable in generating physically feasible low-level action sequences, making it necessary to combine LLMs with conventional motion planning and control modules. Despite these advances, LLM-based multi-agent TAMP systems still face significant scalability and efficiency challenges, such as the effects of the design of communication architectures among multiple LLM agents on planning efficiency and latency (Park et al. 2026; Chen et al. 2024). Meanwhile, recent advances in prompt engineering and memory augmentation techniques, such as retrieval-augmented generation (Wang et al.

2025; Sun et al. 2025), structured memory integration (Yuen et al. 2026) and fine-tuning (Li et al. 2025), have demonstrated strong potential in improving planning reliability. These technologies have been proved effective in reducing hallucinations, and increasing task success rates in single-agent settings (Lin et al. 2025). For example, Sun et al. (2025) applied a retrieval-augmented hierarchical in-context reinforcement learning framework, achieving performance improvements of 9%, 42%, and 10% over the SOTA baseline across five execution episodes. These developments suggest promising opportunities for enhancing multi-agent TAMP systems.

4.3 Decentralized systems and adaptive planning

Adaptive planning, as discussed in Sect. 3.3, involves the re-planning of tasks, the adaptation of motion paths, or in some cases, both. The emergence of adaptive planning is primarily driven by uncertainty challenges in real-world TAMP scenarios, such as task variations, partially observed environments or unpredictable external disturbances. Under such conditions, fully offline planning, pre-computing all action sequences in advance, is infeasible. Consequently, online planning plays a critical role, enabling agents to perform real-time computation during task execution. This allows agents to handle dynamic obstacles, allocate newly arrived tasks quickly and adjust execution strategies based on updated environmental information, thereby improving planning robustness and success rates. When scaling to multi-agent systems, adaptive planning becomes significantly more challenging. Such adaptability requires higher computational capability and faster information exchange among agents. Consequently, decentralized systems have gained significant attention for their ability to distribute the computational burden and decision-making process across agents to eliminate the need to wait for centralized data analysis and instruction dissemination. Therefore, for multi-agent adaptive TAMP problems, the integration of online planning with decentralized systems has emerged as a prominent trend. Recent studies have explored various frameworks that address adaptive planning problems under decentralized systems have been summarized in Table 5.

Table 5 Decentralized systems and adaptive planning

References	(Agrawal et al. 2022)	(Liu et al. 2021)	(Xu et al. 2024)	(Chen et al. 2021)
Planning problem	MDP	GBA	CO	MDP
Task planning algorithms	RL and heuristics	Probabilistic Traffic Model and GCG	LRCA	Max-sum
Motion planning algorithms	A* and ORCA	CBS and roll planning	ORCA and GOS and improved Dubin curve	forward dynamic programming
Integrated structure	Interleaved	Interleaved	Task first	Task first
Level of decentralization	Task planning centralized but motion planning decentralized	Task planning centralized but motion planning decentralized	Fully decentralized	Fully decentralized
Planner adaptability	Adaptive	Adaptive	Adaptive	Adaptive
Online planning function	Avoid collision and navigate	Avoid collision and deadlock	Optimize task allocation and avoid collision	Optimize task allocation and avoid collision

LRCA: Lazy-based Review Consensus Algorithm, GOS: Guidance Point Strategy

To solve MAPD problem in warehouse environments, Agrawal et al. (2022) proposed a double level architecture. The high level is designed for task allocation and the low level is designed for MAPF. One of the highlights is the development of the decentralized ORCA in the low level. In their motion planning module of TAMP, the standard A* search algorithm is used to generate static obstacle-avoidance paths, while dynamic obstacle avoidance is achieved through the decentralized ORCA. They prove that the decentralized scheme can offer several advantages, including the ability to handle dynamic obstacles and unknown environments, as well as scalability to a large number of agents. In this decentralized motion planning module, each agent utilizes knowledge of its local environment, the positions, velocities, and dimensions of its neighboring agents, to make independent navigation decisions.

Similar hierarchical planning architectures that combine high-level centralized task planning with low-level decentralized motion planning have also been proposed in Liu et al. (2021). In the low-level path coordination module, Liu et al. (2021) adopted a decentralized structure which contains robot state surveillance, uncertainty handling and CBS-based cooperative path planning modules. This module is responsible for generating fine-grained road-level paths from the local sector paths assigned by the high-level planner. It also incorporates rolling planning strategy to ensure that the system can respond in real time to variations in the number of robots and execution uncertainties.

Xu et al. (2024) proposed a hierarchical full-decentralized framework to address the TAPF problem for multiple non-holonomic vehicles operating in dense environments. At the high level, they introduced a novel decentralized algorithm, the Lazy-based Review Consensus Algorithm (LRCA), to perform task allocation. LRCA provides near-optimal task assignment results by reducing intersections between vehicle–task and task–task paths. At the low level, they proposed a distributed motion planning algorithm called the Guidance Point Strategy (GOS), which integrates an improved Dubin curve with the ORCA model to resolve vehicle deadlocks and generate feasible velocities and paths. Extensive simulation experiments in large-scale scenarios demonstrated that, compared with centralized approaches, decentralized approaches generally do not provide globally optimal solutions but can guarantee computational efficiency, especially in highly dynamic environments.

Chen et al. (2021) proposed a fully decentralized Task and Path Planning framework for multi-robot systems. Each task is modeled as a MDP or a Mixed Observed Markov Decision Process (MOMDP). The task allocation module maximizes the expected pure reward by formulating the Markov model as a factor graph, which allows decentralized optimization using the max-sum algorithm. Each robot follows the optimal policy derived from its Markov model, while a localized forward dynamic programming scheme resolves conflicts and avoids collisions. Robots share local observations to maintain a common belief over unobserved states, enabling coordinated decision-making. This framework integrates probabilistic reasoning and decentralized optimization, offering a scalable and robust solution for decentralized multi-robot TAMP.

As shown by above research, decentralized systems are generally more scalable than centralized systems when implementing real-time planning in multi-agent settings. However, due to their reliance on local information exchange and the absence of a global perspective, decentralized systems tend to produce suboptimal solutions. Nevertheless, decentralized online planning is still a widely adopted strategy despite its theoretical suboptimality. Because in the context of real-time dynamic TAMP problems, where rapid

Table 6 Reinforcement learning strategy

References	(Agrawal et al. 2022)	(Li et al. 2024)	(Xu et al. 2023)
Planning problem	MDP	CO	POMDP
Task planning algorithms	RL and heuristics	RL and MPC and GNN	MAPPO
Motion planning algorithms	A* and ORCA	RL and MPC and GNN	MAPPO
integrated structure	Interleaved	Interleaved	Interleaved
Communication structure	Decentralized	Decentralized	Decentralized
Planner adaptability	Adaptive	Adaptive	Adaptive
RL reward function	Total time taken to do nothing	Maximize the cumulative reward	Task logic correctness, minimize idle time, minimize path length, collision avoidance

POMDP: partially observable Markov decision process,
MAPPO: multi-agent proximal policy optimization

reaction to environmental changes is critical, decentralized systems exhibit irreplaceable significant advantages. To balance decentralization and global optimality, mainstream solutions include hierarchical planning, where at the high level allocate tasks centrally while at the low level realize dynamic path planning decentrally. Representative examples include (Agrawal et al. 2022; Liu et al. 2021; Chen et al. 2024; Zhao et al. 2024). Another solution is introducing consensus-based protocols, allowing agents to exchange bids through limited local communication to approximate the global optimal task allocation within a decentralized framework. Xu et al. (2024); Peti et al. (2023); Wang et al. (2022) serve as representative examples. Overall, whether decentralized systems are the optimal paradigm for online adaptive planning remains an open question. Further research is required to explore how multi-agent adaptive TAMP can achieve improved scalability, real-time responsiveness, and near-global optimality simultaneously.

4.4 Reinforcement learning strategy

RL has the advantage of robust learning capabilities and optimizing complex and long-term decision-making processes, so it has emerged as a promising solution for multi-agent TAMP problems. Multi-agent reinforcement learning (MARL) refers to a paradigm in which multiple agents learn through interactions with both the environment and other agents, aiming to maximize individual or collective rewards while accomplishing their tasks. Representative works in this area are summarized in Table 6.

As mentioned in Sect. 4.3, Agrawal et al. (2022) proposed a double level architecture. The high level is designed to formulate task allocation problems into MDP and solve it through RL. The goal of the RL is minimizing the total time taken to do nothing, which is generated by the low level. The reward function incorporates both congestion and collision

feedback. Specifically, the RL reward, total travel delay, is generated from the feedback of low-level navigation algorithm and used to train the policy for task allocation.

Except used in task planning module of multi-agent TAMP, RL can be also used in path planning module. For instance, Li et al. (2024) proposed a new model which integrates deep reinforcement learning (DRL), MPC, and GNN to optimize path planning and task allocation in multi-robot systems. DRL is responsible for learning the policy and optimizing decision-making strategies according to the dynamics of the environment. It enables individual robots to make optimal decisions that adapt to changing task requirements and environmental conditions. MPC is used to fine-tune the control actions. Specifically, the integration of DRL with MPC allows for learning-based strategic decision-making while ensuring real-time control adjustments. GNN is designed to deal with robot communication and cooperation. This unique combination thus improves the overall performance of multi-robot systems.

Xu et al. (2023) tried to adopt DRL algorithms into onsite construction robotic TAMP. The authors formulated the problem as a partially observable Markov decision process (POMDP). The goal is to finish the path-finding process and arrive at the waypoint on an optimal schedule to start the construction activities with the shortest time span and distances navigated. The authors proposed a systematic framework, where multi-agent proximal policy optimization (MAPPO) algorithm is adopted for MATA and MAPF adapts it to complicated TAMP. MAPPO is a policy gradient-based actor-critic algorithm which belongs to MARL algorithms. In this research, the policy and value networks are trained under a centralized training and decentralized execution (CTDE) framework in which each robot has a copy of the policy and the value networks. Meanwhile, they are trained on the experiences collected by all the robots simultaneously. The reward function incorporates feedback from both MATA and MAPF performance. Specifically, the reward includes penalties for incorrect task assignment logic, excessive idle time, long travel distances, and collisions. By integrating these objectives into a unified reward function, the framework enables coordinated multi-robot behavior that improves overall efficiency and reduces task completion time.

CTDE has emerged as a dominant framework in MARL, primarily due to its ability to mitigate the non-stationarity problem inherent in multi-agent environments (Mianji et al. 2024). Non-stationarity arises because each agent continuously updates its policy during training, causing the state transition probabilities and reward functions change from the perspective of any individual agent, thereby violating the MDP assumption (Lan 2025). The CTDE framework addresses this challenge by introducing centralized critics or global state information during training, allowing agents to account for the policies and interactions of others (Wang et al. 2022). As a result, the environment becomes relatively more stable during training, significantly improving convergence and learning efficiency. For example, Feng et al. (2024) proposed the Hierarchical Consensus-based Multi-Agent Reinforcement Learning (HC-MARL) framework based on the CTDE framework, to address the limitation of lacking global signals. HC-MARL leverages contrastive learning to enable agents to learn globally consistent representations from local observations, allowing them to achieve implicit consensus without explicit communication.

Despite its advantages, CTDE faces several fundamental limitations. One key challenge lies in credit assignment under global reward settings, where accurately quantifying each agent's contribution to the overall outcome remains difficult (Wei et al. 2025). Additionally, as the number of agents increases, the joint action space grows exponentially, lead-

ing to significant computational overhead for centralized critics and limiting scalability to large-scale systems (Li et al. 2025). Recent studies Zhou et al. (2025) further indicate that existing CTDE frameworks may not fully exploit available global information, thereby constraining perception and exploration efficiency. To address these limitations, recent research has explored both architectural innovations and algorithmic improvements. For instance, Guo et al. (2025) proposed a novel MARL framework named Scalable and Heterogeneous Proximal Policy Optimization (SHPPPO), integrating heterogeneity into parameter-shared PPO-based MARL networks to achieve zero-shot scalable multi-agent collaboration. Similarly, Zhou et al. introduced a novel Centralized Advising and Decentralized Pruning (CADP) framework that not only enables an efficacious message exchange among agents during training but also guarantees the independent policies for execution. Other efforts have focused on improving learning algorithms within the CTDE paradigm, such as the multi-task double deep Q-network approach proposed by Zhu et al. (2023), and enhanced policy gradient methods introduced by Fu et al. (2022), including independent, individual, and autoregressive policy variants. Dai et al. (2025) introduced a constrained flashforward mechanism and an attention mechanism that reasons about both short-term cooperation and long-term scheduling dependency to improve the efficiency of RL-based heterogeneous MATA. However, most of these advances have primarily been validated in simulated environments, and their application to real-world multi-agent TAMP systems remain limited. Bridging this gap represents an important future research direction, particularly in practical domains such as automated warehousing and logistics, intelligent manufacturing and flexible assembly, and multi-robot search, rescue, and monitoring, where scalable, adaptive, and robust multi-agent coordination is essential.

5 Future trends

5.1 VLM-based planning

Vision Language Models (VLMs) can be regarded as an extension of Large Language Models (LLMs), capable of describing problems both textually and visually through one or more images. To enhance the flexibility of robotic planning in interacting with dynamic environments, integrating external vision models can effectively improve environmental perception. Building upon this idea, VLMs have recently become one of the most active research directions.

Numerous studies have leveraged VLMs to assist robots in solving planning related challenges. Yang et al. (2025) proposed VLM-TAMP, a hierarchical planning algorithm that leverages a VLM to generate both semantically meaningful and horizon reducing intermediate sub-goals, which are then solved by a TAMP algorithm. If one sub-goal fails, the VLM will generate a new sub-goal based on the current state of the world and objects that are collided in simulation during the last failed planning process. To deal with last-mile delivery problem, Wang et al. (2025) utilized multi-level Dijkstra algorithm to generate optimal paths between geographic coordinates, and used a VLM to analyze fine-grained details such as the target entrance in order to improve global localization and update the map. Mei et al. (2024) proposed ReplanVLM, a paradigm for task planning developed based on GPT-4V. The framework introduces both internal and external error correction mechanisms to enable

more adaptive planning in long-horizon tasks. Akiyama et al. (2024) compared different language-based foundation models in terms of their reasoning capabilities for task-level planning.

However, most of these efforts focus on single-robot systems or are limited to path planning and navigation tasks. Based on this observation, VLMs hold significant potential for future research in multi-agent TAMP, where reasoning, coordination, and communication among agents remain major challenges. By integrating visual understanding with linguistic reasoning, VLMs could enable robots to interpret high-level semantic instructions, share situational awareness, and negotiate task decisions collaboratively. Furthermore, when combined with decentralized control and adaptive planning strategies, VLM based frameworks may provide a unified approach for perception, planning, and communication across heterogeneous agents.

Despite these promising capabilities, several fundamental challenges must be addressed before VLMs can be reliably applied to multi-agent systems. First, multimodal alignment and grounding remain critical issues, as VLMs must accurately associate high-level linguistic concepts with spatially and temporally grounded physical entities (Tang et al. 2026). In multi-robot environments, even when different agents share the same visual scene, inconsistent observations, occlusions, and partial viewpoints may lead to misinterpretations of object states or task contexts, resulting in incorrect sub-goal generation or coordination failures. A common solution is the Joint Alignment Tuning framework (Zhu et al. 2025), which enables planning and execution agents to iteratively adapt to each other's behavioral patterns through interaction, thereby aligning the generated sub-goal sequences with the agents' actual execution capabilities.

Second, real-time performance and reliability are essential requirements for robotic systems operating in dynamic environments. For example, the inference latency of VLMs (e.g., GPT-4V, LLaVA) typically ranges from hundreds of milliseconds to seconds, whereas low-level motion planning and control usually operate at frequencies of 10–100 Hz (Jiang et al. 2026; Williams et al. 2026). A common solution is to adopt a hierarchical architecture, where the VLM is responsible only for high-level topological planning or sub-goal generation, while lightweight low-level controllers handle execution (Yang et al. 2025). In addition, when multiple agents simultaneously invoke large-scale VLM APIs or share GPU memory resources, severe resource contention and bandwidth bottlenecks may arise, leading to inconsistent system responsiveness. One solution is to introduce small language models (SLMs) deployed on edge devices to replace general-purpose large models, trading a marginal reduction in perception accuracy for significantly faster inference speed. For example, Ni et al. (2025) developed SwiftVLA, which achieves performance comparable to VLA models that are seven times larger, while delivering an $18\times$ speedup and a $12\times$ reduction in memory usage on edge devices. Closed-loop re-planning mechanisms (Lou et al. 2025; Budzianowski et al. 2025) have also been proposed to improve system responsiveness and adaptability. To address reliability problems caused by perceptual hallucinations and environmental disturbances, frameworks such as ReplanVLM (Mei et al. 2024) continuously compare predicted states with real-time visual observations and perform immediate re-planning when deviations are detected. This framework can gain an average error detection rate of 95.7%, with an average error correction rate of 80%.

Third, computational cost and scalability pose significant barriers to practical deployment. SOTA VLMs typically require substantial computational resources and exhibit high

inference latency (Gao et al. 2025), which becomes particularly problematic in multi-agent settings where frequent perception updates, inter-agent communication, and iterative re-planning are necessary (Li et al. 2025). This challenge is further amplified as the number of agents and task complexity increase.

VLM offer a promising future for enhancing multi-agent TAMP by enabling semantic-level perception, high-level reasoning, and natural language-guided coordination in complex and dynamic environments. With continued advances in multi-modal alignment, hierarchical architectures, and real-time feedback mechanisms, VLM-integrated TAMP systems have the potential to significantly improve adaptability, scalability, and autonomy in real-world multi-robot applications.

5.2 Long-horizon task planning

Long-horizon task planning now is the most popular research topic in robot task planning. Long-horizon tasks, usually characterized by complex subtask dependencies, present a significant challenge in real-world planning issue. In real-life applications, robotic planning, particularly manipulator planning, faces the fundamental challenge of searching within an extremely large planning space, which grows exponentially with the length of the task. Moreover, real-world environments often involve dynamic interactions, requiring not only accurate perception of environmental states but also the ability to predict the causal effects of the robot's actions on the environment. While TAMP planners typically require complete knowledge of the underlying system states and the kinodynamic models of the environment, which significantly limits their applicability in real-world scenarios. To address these challenges, language based approaches, e.g. VLMs (Yang et al. 2025), (Vision Language Action Models)VLAs (Zhao et al. 2025; Fan et al. 2025), LLMs (Guo et al. 2025; Zhou et al. 2024), and diffusion model (Chen et al. 2024; Mishra et al. 2023) based approaches have become promising solutions in recent research.

The successful and representative application of LLM and VLM based approaches have been discussed in Sects. 4.2 and 5.1. As mentioned above, VLM can be seen as an extension of LLM. Building on this, VLAs further extend VLMs by leveraging their rich perceptual and reasoning capabilities to map natural language instructions and visual observations into executable robot actions. Zhao et al. (2025) observed that most mainstream VLAs adopt a direct end-to-end input-output mapping paradigm, which lacks intermediate reasoning and therefore performs poorly on complex manipulation tasks. To address this limitation, they proposed CoT-VLA, which introduces visual chain-of-thought reasoning into VLA models. Specifically, the framework first generates subgoal images as intermediate reasoning steps, and then produces short action sequences that achieve these visual subgoals. This approach significantly improves the performance of VLA models on long-horizon and multi-step manipulation tasks. Fan et al. (2025) proposed Long-VLA, an end-to-end VLA model specifically designed for long-horizon robotic tasks. The model automatically identifies the moving phase and the interaction phase within each subtask, and emphasizes different types of perceptual cues in each phase accordingly. This phase-aware design improves the compatibility between subtasks and enhances the model's overall ability to handle long-duration manipulation tasks.

Despite Language based approaches, diffusion models are another feasible solutions. Diffusion models are particularly effective at representing multimodal distributions. Chen

et al. (2024) proposed a hierarchical diffuser to address the computational and generalization challenges of diffusion-based planning in long-horizon tasks. In this framework, the high-level module performs “jumpy” planning by generating sub-goals over extended temporal horizons, instead of rolling out actions at every single timestep. Guided by these high-level sub-goals, the low-level module then conduct the fine-grained path refinement to produce executable motion plans. Mishra et al. (2023) proposed Generative Skill Chaining, a long-horizon task planning framework that leverages diffusion skill models to probabilistically compose skills. The framework performs parallel sampling over all skill distributions during inference, enabling efficient generation of long action sequences while supporting geometric constraint handling and real-time re-planning.

At present, all these research mentioned above are performed in single manipulator system but we consider that VLA-based approaches will have a promising potential in multi-robot systems after solving the communication and coordination problems in the future.

5.3 Theory-based planning

Theory-based planning refers to planning approaches based on rigorous mathematical models, physical laws, or formal decision-making theories, in contrast to purely data-driven or end-to-end learning methods (Hu et al. 2025). Common theoretical foundations include network flow theory, game theory, belief space theory, and control theory.

Network flow theory formulates planning as a constrained flow optimization problem that ensures collision avoidance or efficient task allocation. To deal with the problem of allocating goals to multiple robots with complex dynamics while computing obstacle-free motions to reach those goals, Salvado et al. (2021) used an graph-based method for decomposing the problem into sub-problems. In particular, they reduced the problem to a Minimum-Cost Max-Flow problem whose solution is used by a multi-robot motion planner that does not impose restrictive assumptions on robot kinodynamics or on the environment. Liu et al. (2024) decomposed the environment into a flow-based network structure, enabling the efficient allocation of robots to paths based on real-time congestion levels.

Game theory has been widely applied in the field of autonomous vehicles (AVs), particularly in decision-making (Liu et al. 2022; Lopez et al. 2022) and motion planning (Yuan et al. 2023; Burger et al. 2022). For example, Huang et al. (2023) tackled the interaction prediction problem in real-world AVs operating by formulating it with hierarchical game theory and proposing the GameFormer model for its implementation. Yan et al. (2023) proposed a game theoretic trajectory planning framework, including non-cooperative games between AVs and human-driven vehicles, and partial cooperative games between lane-changing AV and other AVs. They also proposed a longitudinal speed adjustment strategy based on the partial cooperative game theory. To develop a new method for multi-agent planning involving human drivers and AVs in unsignaled intersections, roundabouts, and during merging, Chandra and Manocha (2022) proposed a new auction, GamePlan, based on game theory. It can determine the optimal action for each agent based on their driving style and prevents collisions and deadlocks.

Belief space theory addresses planning under uncertainty by representing the system state as a probability distribution, and is commonly applied in autonomous navigation and exploration tasks where sensing and localization are imperfect. Jiang (2024) proposed framework that formulates multi-robot optimal control as probabilistic inference over graphical mod-

els and leverages belief propagation to achieve inference via distributed computation. This framework has been validated in a multi-robot formation navigation problem. Patwardhan and Davison (2024) formulated the distributed coordinated robots exploration task problem as a single factor graph with multiple connected layers representing each aspect and adopted Gaussian Belief Propagation (GBP) as the inference mechanism.

Control theory focuses on generating dynamically feasible and stable trajectories, and is widely used in motion planning, trajectory tracking, and coordinated control of robots. Particularly MPC has proven useful in motion planning. Askari et al. (2025) proposed the idea of model predictive inferential control (MPIC), which seeks to infer the best control decisions from the control objectives and constraints, to achieve MPC-based motion planning when a neural state-space model represents the vehicle dynamics. Hu et al. (2021) designed a real-time nonlinear model predictive control based motion planner (NMPC-MP) for teleoperated manipulation to generate smooth and reliable trajectories.

Theory-based planning offers several fundamental advantages, including formal correctness guarantees, strong interpretability, and the ability to explicitly model system dynamics, constraints, and objectives within a mathematically rigorous framework (Hu et al. 2025). In particular, game-theoretic (Liu and Li 2025; Fei et al. 2025) and control-theoretic (Wang et al. 2025; Costa and Ogren 2025) approaches enable agents to anticipate the behaviors of others and optimize their own actions accordingly, which improves safety, robustness, and coordination efficiency in highly interactive and uncertain environments. Theory-based planning offers several fundamental advantages, including formal correctness guarantees, strong interpretability, and the ability to explicitly model system dynamics, constraints, and objectives within a mathematically rigorous framework (Antonyshyn et al. 2023). Benefiting from these advantages, theory-based planning has gradually been extended to Multi-agent TAMP. Xu et al. (2025) proposed a game-theoretic framework to integrate task allocation, path planning and formation control for multi-agent systems. In this framework, an anonymous hedonic game (AHG) is established for allocating tasks according to agents' preferences, while a Stackelberg differential graphical game is nested within the AHG for path planning and controlling agents' formation. The proposed framework makes the strategies to be designed in a formation 'control-path planning-task allocation' closed-loop form and the performance of each phase to be bidirectionally adjustable. Meanwhile, they developed an integrated game algorithm to optimize the parameters of the strategies.

However, theory-based planning methods remain relatively underexplored in the context of Multi-agent TAMP. Existing studies are largely concentrated in AVs and typically address only specific subproblems, such as MATA or MAPF, rather than the full integration of TAMP. This limitation can be attributed to several inherent challenges of theory-based planning. First, reliance on simplified models and restrictive assumptions limits their ability to capture the combinatorial and hybrid nature of Multi-agent TAMP (Xie et al. 2025). Second, scalability remains a major challenge due to the exponential growth of state and strategy spaces (Ma et al. 2022). As a result, their applicability to large-scale Multi-agent TAMP remains limited.

Addressing these challenges represents an important research direction. Future work could focus on developing scalable theoretical frameworks that incorporate hierarchical decomposition, distributed optimization, and abstraction techniques to reduce computational complexity while preserving theoretical guarantees. In addition, integrating theory-based planning with learning-based methods may enhance adaptability to uncertainty and

enable more efficient approximation of complex decision spaces. By overcoming these limitations while retaining their inherent advantages in rigor and interpretability, theory-based planning has the potential to play a more prominent role in broader Multi-agent TAMP applications, such as search and rescue, warehouse automation, cooperative robotic exploration and so on.

5.4 Data privacy

With the rapid development of mobile internet and the sharing economy, spatial crowd-sourcing has become increasingly prevalent. However, privacy concerns associated with users' geographic locations and moving paths have also become more prominent. As a result, data privacy issues in online task allocation have attracted growing attention from the research community. Zhang et al. (2024) observed the need for privacy preservation in online task and path planning for spatial crowd-sensing on untrusted platforms. From a data security perspective, the authors did not adopt classical task allocation and path planning algorithms, as these algorithms require comparing plain text distance values, which is infeasible in the encrypted domain where all data in POTA are cipher texts. They proposed POTA, the first privacy-preserving online multi-task allocation scheme with path planning, which minimizes the total moving distance of crowd-sensing workers without introducing additional noise. POTA cryptographically implements an extended minimum-cost flow model that represents the encrypted data of workers and tasks in a graph and subsequently produces optimized routing. Specifically, POTA computes encrypted distances using homomorphic encryption, determines the shorter distance through a secure comparison protocol, and achieves globally optimal matching under the Minimum Cost Flow framework.

In addition, with the explosive development of LLMs, Tool-Using LLMs have recently received increasing attention due to their ability to interact seamlessly with various external tools and APIs. However, such capability also introduces hidden risks that private user information transmitted to the LLM may be unintentionally exposed to the external tools it invokes. To address this issue, Zhang et al. (2024) proposed PrivacyAsst, a privacy preserving framework specifically designed for tool-using LLM agents. While in multi-agent TAMP, LLMs can also serve as autonomous agent driver, and the multi-agent TAMP process shares conceptual similarities with tool-using LLM settings. Therefore, PrivacyAsst and similar privacy protection frameworks hold strong potential for enhancing privacy guarantees for LLM-powered autonomous agents in future multi-agent TAMP research. For example, by executing tasks over ciphertext to ensure computational security, or by adopting shuffling-based mechanisms to support private input handling in task-agnostic scenarios.

While the mobility, perception, and interaction capabilities of autonomous robots bring tremendous convenience to daily life, they also introduce significant privacy risks. Shome et al. (2023) conducted a systematic review to address this research gap in privacy issues within robotic motion planning. They formulated privacy-aware motion planning as a cost function based problem, and through simulation experiments and navigation case studies, demonstrated that privacy vulnerabilities can arise directly from the way such cost functions are defined. Their findings highlight the need for more in-depth investigations into privacy preserving autonomous robots planning in future research.

6 Conclusion

This survey provided a structured review of recent progress in multi-agent TAMP. Based on 60 representative papers from 2021 to 2025, we proposed a unified taxonomy from three perspectives: integrated structures, communication systems, and planner adaptability. Through this taxonomy, we summarized and analyzed the dominant design choices in existing literature. Our analysis shows that task-first centralized fixed-plan frameworks remain the most widely adopted strategy, followed closely by interleaved centralized fixed-plan frameworks, with comparable proportions. In recent years, decentralized and adaptive approaches have attracted increasing attention due to their superior scalability and robustness in dynamic and uncertain environments.

We further reviewed emerging trends in multi-agent TAMP, including planning structure design, decentralized and online re-planning mechanisms, LLM-based approaches, and reinforcement learning strategies. Looking ahead, we anticipate that future research will focus on integrating VLMs, addressing long-horizon tasks, advancing theory-based planning methods, and incorporating privacy-preserving computation techniques. We hope this survey provides a useful reference for understanding the current state of the art and inspires new research directions toward more scalable, adaptive, and intelligent cooperative robotic systems.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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